

# Superstar Spillovers in the WNBA:

## The Effect of Caitlin Clark Exposure on Players' Digital Attention

Applied Econometrics (EC4102) Mid-Term Research Proposal

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### 1 Motivation

In many production settings such as firms, research labs and sports teams, a small number of highly productive individuals work with many other workers. This raises an important question in labor economics. Can a superstar also increase the visibility, productivity or value of the people around them?

This question is important because firms may want to structure teams and allocate marketing and promotion resources around these individuals. It can also affect the compensation and market value of workers who are associated with a superstar.

Professional sports provide a useful setting to study this question because outcomes such as attention and revenue can be measured at the individual level and on a daily basis. Caitlin Clark's entry into the WNBA in 2024 provides a useful case for this. Her presence has been associated with increases in ticket sales, television viewership and local spending for both her own team and the teams she plays against.

However, it is not clear whether this effect also exists at the individual player level. In particular, we want to know whether players who share the court with Clark receive more public attention, either as teammates or as opponents.

We therefore focus on the following question:

Does sharing the court with Caitlin Clark increase an individual player's public attention, after controlling for the player's own performance?

### 2 Literature Survey

Rosen (1981) provides the theoretical basis for the idea of superstar effects. Small differences in talent can lead to large differences in earnings and visibility when technology allows a large number of people to consume the output of a highly talented individual.

Humphreys and Johnson (2020) study the effect of superstar players on NBA game attendance. They use a team-game panel with team and season fixed effects, a star player indicator and controls for opponent quality. This provides a useful framework for studying superstar effects in sports.

Stephenson (2024) studies the economic impact of Caitlin Clark's college career using daily hotel occupancy data around Iowa City. The study uses date and event fixed effects. The estimated effect was smaller in Clark's senior year than in her junior year. The paper suggests that this may partly be because people substituted from men's games to women's games instead of creating completely new economic activity.

A recent paper, "Superstar Effects in the WNBA" (2026), studies WNBA attendance from 1997 to 2024 and looks at 27 identified star players. The paper finds that Clark's effect on home and away attendance is larger than that of previous WNBA stars.

Zimmer, Snyder and Zimmer (2025) also study WNBA attendance and examine whether the increase in attendance in 2024 can be explained by the Clark and Reese rookie class or by broader league growth.

#### The Gap

The existing studies mainly look at outcomes such as attendance, hotel occupancy and league revenue. They do not directly study whether individual players become more visible when they share the court with a superstar.

Our study focuses on this individual level effect. We ask whether a particular teammate or opponent receives more public attention on days when they play with or against Clark, while controlling for their own performance.

### 3 The Research Question and the Ideal Experiment

#### 3.1 Statement of the Research Question

Our main research question is:

Does sharing the court with Caitlin Clark, either as a teammate or as an opposing player, increase a WNBA player’s public attention?

We measure public attention using daily Wikipedia pageviews of the player’s own article.

We also want to study whether the effect is different for teammates and opponents. Teammates share the same team and broadcast exposure with Clark, while opponents only play against her. We will also check whether any increase in attention continues into the next day.

### 3.2 The Experimental Ideal

The unit of observation is an active WNBA player other than Clark herself, observed on each day during the 2023–2025 seasons. The 2023 season is used as the pre-Clark period.

The treatment variable is defined as:

$$D_{it} = \begin{cases} 1, & \text{if player } i \text{ plays in a game featuring Clark on day } t, \\ 0, & \text{otherwise.} \end{cases}$$

We separate this treatment into two variables:

$$TeammateClark_{it}$$

and

$$RivalClark_{it}.$$

In an ideal experiment, whether a player plays with or against Clark would be randomly assigned. Other factors affecting attention, such as the player’s own performance, national television coverage and opponent quality, would be held fixed.

The outcome variable is:

$$Y_{it} = \log(1 + \text{Wikipedia pageviews}_{it})$$

for player  $i$  on day  $t$ .

We will also use pageviews on day  $t + 1$  as a secondary outcome to see whether the effect continues after the game.

Under random treatment assignment, the potential outcomes would be independent of treatment:

$$(Y_{it}^1, Y_{it}^0) \perp D_{it}.$$

In that case,

$$E[Y_{it} \mid D_{it} = 1] - E[Y_{it} \mid D_{it} = 0]$$

would give the average causal effect of playing with or against Clark.

However, treatment is not actually randomly assigned in our setting. The league schedule and television coverage are determined before the games and can depend on team and market characteristics. This creates a problem for identifying the causal effect.

## 4 Data Description and Empirical Strategy

The analysis uses a player-date panel constructed from two main data sources. WNBA game and player statistics are obtained from SportsDataverse, while daily Wikipedia pageviews are obtained from the Wikimedia REST API. The WNBA data contain the game ID, date, teams, opponent, and player-level statistics such as minutes, points, assists, and rebounds. The schedule and player files are merged using the game ID, which allows us to identify the players who were teammates of Caitlin Clark and those who faced her.

The treatment is defined at the player-date level based on whether a player shared a game with Caitlin Clark, either as a teammate or an opponent. Wikipedia pageviews are then merged by player and date to measure daily public attention. This gives a panel where each observation contains the player’s game performance, Clark’s presence in the game, and the player’s daily Wikipedia pageviews.

### 4.1 Source and Access

The data we plan to use are publicly available.

- Game logs, box scores, rosters and broadcast information will be collected through SportsDataverse.
- Daily Wikipedia pageviews will be obtained from the Wikimedia Pageviews REST API.

- Google Trends and Social Blade data will be used as additional outcomes for robustness checks.

## 4.2 Structure

The data will be organised as a player-day panel. We expect around 140–150 active players across the three seasons. The final number of observations will depend on the availability of Wikipedia pages and other variables.

## 4.3 Key Variables

The treatment variables are:

$$TeammateClark_{it}$$

and

$$RivalClark_{it}.$$

The main control variables are the player’s own performance and characteristics of the game. These include:

- points,
- rebounds,
- assists,
- national television coverage,
- home or away status,
- opponent win percentage.

We will also use the player’s pre-sample average pageviews as a measure of baseline popularity.

There are some measurement issues with the data. Wikipedia pageviews can contain automated traffic, although the user access filter reduces this issue. Pageviews can also increase because of events unrelated to Clark, such as injuries, trades or other news.

Players without an existing Wikipedia article will not be included in the sample. This is a limitation of the study because these players may be different from players who already have Wikipedia pages.

## 4.4 Descriptive Statistics

We will first compare average pageviews on Clark game days and non-Clark game days.

The comparison will be made separately for teammates and opponents. We will also perform the same comparison for the 2023 season as a placebo check.

Finally, we will plot average pageviews around games involving Clark to see how attention changes before and after the game.

# 5 Identification Strategy

## 5.1 Why OLS is Not Enough

A simple regression would be:

$$Y_{it} = \beta_0 + \beta_1 D_{it} + u_{it}.$$

However, this regression may not give us the causal effect because:

$$\text{Cov}(D_{it}, u_{it}) \neq 0.$$

One reason is the selection of games for national television. Popular teams and players are more likely to be included in nationally televised games. Many Clark games may also receive national television coverage.

This means that some players may already have higher public attention even without Clark. In this case, the treatment variable is related to factors that also affect the outcome.

Another issue is the player’s own performance. A player may perform better when playing against a high-profile opponent. If this happens, the increase in pageviews may come from the player’s own performance rather than simply from playing with or against Clark.

Therefore, treated and untreated player-days may already be different before considering the treatment:

$$E[Y_{it}^0 \mid D_{it} = 1] \neq E[Y_{it}^0 \mid D_{it} = 0].$$

The main confounding path can be written as:

$$D_{it} \leftarrow \text{Player popularity} \rightarrow Y_{it}$$

A simple comparison between treated and untreated observations would not account for this path.

## 5.2 Empirical Strategy

Our main regression is:

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 TeammateClark_{it} + \beta_2 RivalClark_{it} + \beta_3' X_{it} + \epsilon_{it}.$$

Here,  $\alpha_i$  is the player fixed effect and  $\lambda_t$  is the week or season fixed effect.

The vector  $X_{it}$  contains the control variables.

We define the control vector as:

$$X_{it} = \begin{bmatrix} Points_{it} \\ Rebounds_{it} \\ Assists_{it} \\ NationalTV_{it} \\ Home_{it} \\ OpponentWinPct_{it} \end{bmatrix}.$$

The corresponding coefficient vector is:

$$\beta_3 = \begin{bmatrix} \beta_{31} \\ \beta_{32} \\ \beta_{33} \\ \beta_{34} \\ \beta_{35} \\ \beta_{36} \end{bmatrix}.$$

Therefore, the term  $\beta_3' X_{it}$  can be expanded as:

$$\begin{aligned} \beta_3' X_{it} = & \beta_{31} Points_{it} + \beta_{32} Rebounds_{it} + \beta_{33} Assists_{it} \\ & + \beta_{34} NationalTV_{it} + \beta_{35} Home_{it} + \beta_{36} OpponentWinPct_{it}. \end{aligned}$$

So the full regression is:

$$\begin{aligned} Y_{it} = & \alpha_i + \lambda_t + \beta_1 TeammateClark_{it} + \beta_2 RivalClark_{it} \\ & + \beta_{31} Points_{it} + \beta_{32} Rebounds_{it} + \beta_{33} Assists_{it} \\ & + \beta_{34} NationalTV_{it} + \beta_{35} Home_{it} + \beta_{36} OpponentWinPct_{it} + \epsilon_{it}. \end{aligned}$$

The player fixed effect controls for factors that are constant for a player over time, such as their general level of popularity. The time fixed effects control for changes in overall WNBA attention over time.

The national television variable is particularly important because it controls for the fact that some games are more likely to receive attention because they are nationally televised.

Our identifying assumption is that, after controlling for these variables and fixed effects, treatment is independent of the remaining error:

$$D_{it} \perp \epsilon_{it} \mid \alpha_i, \lambda_t, X_{it}.$$

Under this assumption,  $\beta_1$  and  $\beta_2$  can be interpreted as the teammate and rival spillover effects.

## 5.3 Heterogeneity

We will also test whether the teammate and rival effects are different.

The null hypothesis is:

$$H_0 : \beta_1 = \beta_2.$$

The alternative hypothesis is:

$$H_1 : \beta_1 \neq \beta_2.$$

This allows us to check whether being a teammate of Clark has a different effect from simply playing against her.

## 5.4 Placebo Test

As a falsification check, we will use the 2023 season, before Clark entered the WNBA.

We will use a placebo treatment based on a comparably prominent player from 2023. If the model is picking up a general effect of being around a star player rather than a Clark-specific effect, we would expect to see a similar result in the placebo test.

## 6 Inference

Our data contain player-day observations from the WNBA seasons being studied. Since observations from the same player can be related over time, we need to account for this when calculating standard errors.

Our baseline specification will cluster standard errors at the player level.

$$SE(\hat{\beta}) = \text{clustered at the player level.}$$

We will also report a robustness specification where standard errors are clustered at the team-date level.

$$SE(\hat{\beta}) = \text{clustered at the team-date level.}$$

The team-date clustering is useful because players on the same team have the same treatment status on a particular game day.

## 7 References

### References

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